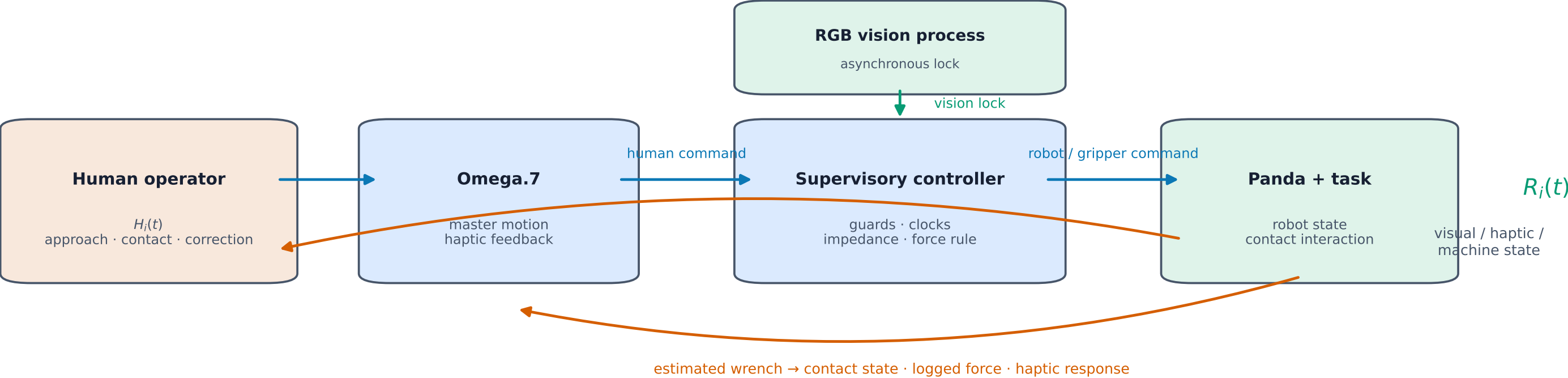
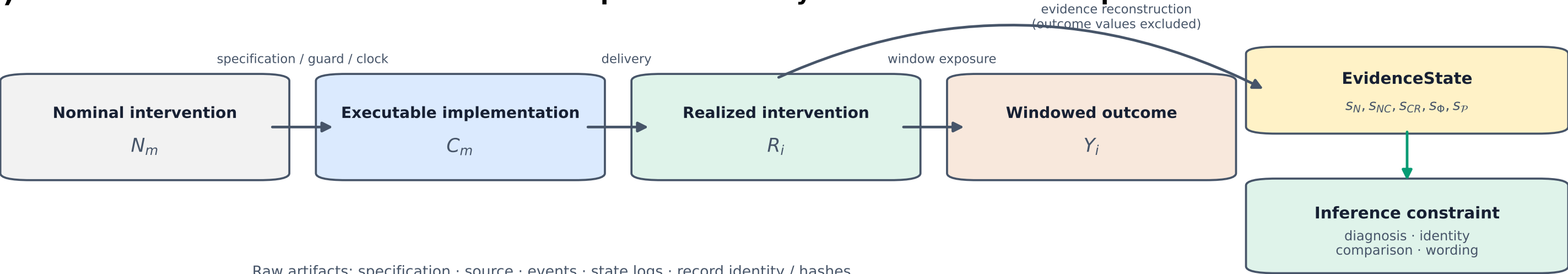


(A) Asynchronous human-machine loop determines intervention delivery and exposure



$$R_i(t) \rightarrow H_i(t + \delta) \rightarrow R_i(t + \delta) \rightarrow Y_i$$

(B) Evidence reconstruction constrains comparison identity before outcome interpretation



Acquisition provenance is an orthogonal prerequisite; identity retained $\neq$ causal identification.